

Ex. No 7: Android Data Extraction Using AFLogical OSE

Course / Lab: Digital Forensics Laboratory

Experiment: Android Data Extraction Using AFLogical OSE

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Overview

AFLogical OSE (Open Source Edition) is a forensic tool designed for logical extraction of data from Android devices. Unlike physical acquisition methods that create bit-for-bit copies of storage, logical extraction pulls structured data such as contacts, call logs, SMS/MMS messages, and device information through the Android content provider framework. This experiment demonstrates the complete workflow of connecting an Android device via ADB, deploying AFLogical OSE, extracting forensic data, and analyzing the results.

Experiment Objectives

1. Set up the forensic environment with ADB (Android Debug Bridge) and verify device connectivity.
 2. Deploy and install the AFLogical OSE APK onto the target Android device.
 3. Perform logical data extraction of contacts, call logs, SMS/MMS, and device metadata.
 4. Transfer extracted `.csv` files to the forensic workstation and analyze the data.
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Software and Tools Required

- **AFLogical OSE** — Downloaded from the official GitHub repository
 - **Android Debug Bridge (ADB)** — Part of Android SDK Platform Tools
 - **Java Runtime Environment (JRE)**
 - An Android device or emulator with **USB Debugging** enabled
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Step 1: Connect the Android Device via ADB

After enabling USB Debugging on the Android device (Settings → Developer Options → USB Debugging), connect it to the forensic workstation via USB. Verify the connection using `adb devices`, then install the AFLogical OSE APK.

```
C:\platform-tools> adb devices
```

```
C:\platform-tools> adb install aflogical-ose.apk
```

Device Connection

```
(kali@student)-[~/Downloads]
$ adb devices
List of devices attached
13703469300003P device
```

Figure 1.1: Connecting device via ADB.

Step 2: Select Data Categories for Extraction

Launch the AFLogical OSE application on the connected Android device. The app presents checkboxes for each extractable data category. Select the forensically relevant categories — **Contacts**, **Call Log**, **SMS/MMS**, and **Device Info** — then tap the **CAPTURE** button to begin extraction.

Data Selection

```
(kali@student)-[~/Downloads]
$ adb shell ls /sdcard/
20241210_35714PMBByGPSMapCamera.jpg
570048714df34e7b9c8190f4e54e64de.mp4
5_6062116729352357748.mp3
9875ed5d9097426cab8fe2b2e5c89aa6.jpg
AirDroid
Android
Beast.2022.1080p.NF.WEB-DL.HIN-Multi.DDP5.1.HEVC-ShiNobi (1).mkv
DCIM
Documents
Download
```

Figure 2.1: Selecting data categories.

Step 3: Transfer and Analyze Extracted Data

After extraction completes, the data is stored as .csv files on the device's storage (typically in /sdcard/aflogical/). Use `adb pull` to transfer the files to the forensic workstation, then open them in a spreadsheet application for analysis.

```
C:\platform-tools> adb pull /sdcard/aflogical/ C:\Forensic_Projects\Cas
```

Data Transfer

```
(kali@student)-[~/Downloads]
$ adb shell content query --uri content://contacts/phones > ~/Desktop/AF

(kali@student)-[~/Downloads]
$ mkdir -p ~/Desktop/AFLogical_Evidence

(kali@student)-[~/Downloads]
$ adb shell content query --uri content://sms > ~/Desktop/AFLogical_Evidence
```

Figure 3.1: Transferring and analyzing extracted data.

III Data Categories Extracted

Category	File Format	Key Fields
Contacts	contacts.csv	Name, Phone Number, Email, Type
Call Log	call_log.csv	Number, Duration, Date, Type (Incoming/Outgoing)
SMS/MMS	sms.csv	Address, Body, Date, Read Status
Device Info	info.csv	Device Model, OS Version, IMEI, Serial Number

Step 4: Cleanup and Evidence Preservation

After confirming all data has been successfully transferred and verified:

```
C:\platform-tools> adb uninstall com.viaforensics.android.aflogical
```

This removes the forensic tool from the target device, ensuring no artifacts are left behind that could compromise the evidence or alert the device owner.

✓ Result

Android forensic data was successfully extracted using AFLogical OSE. The tool was deployed via ADB, logical extraction was performed covering contacts, call logs, SMS/MMS messages, and device metadata, and the resulting `.csv` files were transferred to the forensic workstation for analysis. All extracted data was documented and preserved following chain-of-custody procedures.
